

SHAY MORDECHAI

Offensive Security Researcher · Android Internals · Vulnerability Discovery
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SUMMARY

Offensive Security Researcher specializing in Android internals and malware triage. Recognized for uncovering compiler vulnerabilities (CERT-IL #11265) and publishing research on OS execution boundaries. Focused on multi-stage attack chain analysis across Java, Native, and ARM64.

TECHNICAL SKILLS

- Reverse Engineering · Linux Internals · Android Internals · Frida (DBI) · JADX · IDA Pro · Fuzzing · Native-Hook · Threat Modeling · Memory Models (V8/JIT)
- Python · C/C++ · Java · ARM64 Assembly

PROJECTS & RESEARCH

Detailed write-ups at shay-mordechai.github.io

The Permission Paradox: Binder IPC Native Research (In Progress)

- Structural Investigation:** Reverse-engineering libbinder.so (ARM64) extracted from a physical Samsung S20 device to analyze low-level IPC transaction flows at the /dev/binder ioctl level, investigating the structural boundaries of UI allocation during the application bootstrap.

Android OS Internals & Vulnerability Research (Digital Whisper Publication) | May 2026

<https://shay-mordechai.github.io/blog/he/android-bootstrap-hijacking.html>

- Early Lifecycle Hijacking:** Researched early-stage process entry points (attachBaseContext and ContentProvider.onCreate) to reverse-engineer how the Anatsa Banking Trojan executes unpacked DEX payloads prior to standard Application instantiation.
- Framework Code Inversion:** Developed low-level Frida instrumentation scripts (Spawn & Zygote Hook modes) to intercept LoadedApk.makeApplicationInner, forcing forceDefaultAppClass runtime overrides to neutralize malware payload initialization.
- Dynamic Linker Timing Races:** Defeated strict anti-analysis and stub-packing mechanisms by neutralizing class loading dependencies dynamically via runtime validation manipulation inside BaseDexClassLoader.findClass.

Compiler Security Research — LISP/SBCL Runtime (CERT-IL #11265) | Jul 2025

<https://shay-mordechai.github.io/blog/he/compiler-zero-days-research.html>

- Zero-Day Discovery:** Uncovered 3 unreleased architectural logic flaws in the SBCL compiler's scoping mechanism, enabling complete lexical isolation bypasses during macro expansion (Acknowledged by CERT-IL).
- Automation Harness:** Designed and executed an autonomous, grammar-aware Python semantic fuzzer with an automated feedback loop to bypass strict compiler syntax-checkers and catch state manipulation anomalies.

PROFESSIONAL EXPERIENCE

Independent Security Research & Academic Focus | Jul 2025 – Present

- Conducting advanced self-directed vulnerability research, focusing on Android Internals, compiler isolation mechanisms, and low-level IPC architecture.
- Publishing technical analysis and deep-dives on operating system execution boundaries and binary triage.

Application Security Intern | Israel Tax Authority | Aug 2024 – Jul 2025

- Performed execution path analysis and threat modeling for critical file ingestion pipelines and API hardening workflows, identifying authentication and logic flaws in production financial systems.
- Orchestrated the deployment of SAST tools (Checkmarx) across 15+ environments, enabling proactive mitigation of application-layer vulnerabilities.

EDUCATION

B.Sc. Computer Science | Jerusalem College of Technology | 2021 – 2025 · GPA: 89

Core Focus: Reverse Engineering · Advanced OOP (Java, C++) · Malware Analysis · Network Protocols